

The Grant Winning Formula

The Exact Structure We Used to Win

\$18M+ in NIH, DBT & Wellcome Trust Grants

NIH

DBT / ICMR

Wellcome Trust

\$18M+

grants secured

87%

success rate

120+

proposals

42+

countries

500+

researchers

WHAT'S INSIDE

→ 5-Part Specific Aims Formula

→ Innovation Argument Framework

→ Agency-Specific Rules

→ Resubmission Formula

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THE PROBLEM EVERY FUNDED RESEARCHER KNOWS

Most Grant Rejections Are Not About the Science

We have reviewed hundreds of rejected NIH, DBT, and Wellcome Trust proposals. The research was often excellent. The preliminary data was strong. The investigator was qualified. But the proposals were rejected anyway.

The reason is almost always the same: the science was sound, but the narrative was broken. Reviewers don't fund science — they fund arguments about science, written clearly, structured deliberately, and framed for a study section, not a peer's lab.

"A funded grant is not the best science in the pile. It is the best-argued science in the pile. These are not the same thing." — Senior NIH Study Section Member

The 4 Most Common Reasons Grants Get Rejected

Specific Aims	The punchline is buried in paragraph 3. Reviewers stop reading in paragraph 1 if the 'so what' isn't immediately clear.
Innovation	The section describes what the applicant will do — not why it has never been done before. Innovation must argue, not narrate.
Significance	Written for scientific peers instead of the study section. Reviewers are not always experts in your sub-field — clarity beats jargon every time.
Budget Justification	Treated as an afterthought. Every budget line must be justified scientifically, not financially. Reviewers notice when this section is generic.

What this guide gives you

The exact structural formula behind every proposal in our \$18M+ portfolio — applied across NIH R01/R21, DBT, and Wellcome Trust:

- Part 1 — The 5-Part Specific Aims Formula
- Part 2 — The Innovation Argument Framework
- Part 3 — Significance: Writing for the Reviewer
- Part 4 — Approach: The Logical Progression Structure
- Part 5 — Budget Justification That Doesn't Kill Your Score
- Part 6 — Agency-Specific Rules: NIH · DBT · Wellcome Trust
- Part 7 — The Resubmission Formula

1

The 5-Part Specific Aims Formula

The most critical page in your entire application

The specific aims page is the only page every reviewer reads in full. It determines whether the reviewer approaches your research strategy with enthusiasm or scepticism. One page. Five components.

1

STEP

The Opening Hook — The Problem Statement

Your first sentence must name a specific, consequential, and unsolved problem. Not background. Not literature. A problem written so any reviewer understands why it matters in 10 seconds.

KEY ACTION:

Write your first sentence. Ask: 'Could a non-specialist understand why this matters in 10 seconds?' If not, rewrite it.

2

STEP

The Knowledge Gap — Why This Is Unsolved

After the problem, name the gap precisely. 'Little is known about X' is weak. 'The molecular mechanism by which X causes Y in patients with Z remains uncharacterised' is strong. The gap must feel both real and important.

KEY ACTION:

Write the gap in one sentence. Specific enough that a reviewer can't immediately name a paper that closes it.

3

STEP

Your Hypothesis — The Central Bet

State your central hypothesis in one sentence, in plain language. Testable, falsifiable, directly connected to the gap. Reviewers look for this explicitly. If buried in a paragraph, they are frustrated.

KEY ACTION:

Highlight your hypothesis sentence. If longer than 40 words, it is not a hypothesis — it is a plan. Shorten it.

4

STEP

The Aims — Your 3 Logical Questions

2–3 aims, each answering one testable question. Logically connected but not strictly dependent — if Aim 1 fails, Aims 2 and 3 must still be achievable. State each aim in one sentence.

KEY ACTION:

Write: 'We will [verb] [object] to [demonstrate/determine/establish] [outcome].' Apply this to all 3 aims.

5

STEP

The Payoff — Why This Matters

Close with 2–3 sentences explaining what changes when your aims succeed. For NIH: human health. For Wellcome: global health. For DBT: India's scientific mission. Be explicit — never make reviewers infer significance.

KEY ACTION:

Write the payoff. Answer: 'If every aim succeeds, what is different about science or medicine?' in 2–3 sentences.

2

The Innovation Argument Framework

Why your research has never been done — and must be done now

Innovation is scored by every reviewer on every panel. Most applicants write it last and least carefully — and it shows. Reviewers are explicitly asked: 'Is this innovative?' You must make the argument for them.

The 3-Part Innovation Argument

1. Current state of knowledge

Summarise what the field currently knows or uses. Be specific — cite the leading approach or framework. Keep to 2–3 sentences. This is the baseline against which your innovation is judged.

2. The limitation of what exists

Identify the precise limitation. Not 'understudied' — that is a gap. A limitation is: 'Current methods cannot resolve X because of Y, which means Z remains unanswerable.' Specific, consequential, real.

3. How your approach overcomes it

Explain specifically how your approach overcomes the limitation. Use 'unlike current methods, our approach...' or 'for the first time, this study will...' Phrases that explicitly contrast against the baseline you set in part 1.

"The single most common Innovation failure: the applicant tells the reviewer what they will do, not why no one has done it before. Innovation is a historical argument, not a project description."

3

Significance — Writing for the Reviewer

Arguing impact to a generalist, not a sub-field expert

Most Significance sections are written as if the reviewer is an expert in your exact sub-field. They often are not. Study sections cover broad areas. Write for informed generalists — scientifically literate, but not specialists in your topic.

The Significance Hierarchy**→ Scientific Significance**

How does this advance fundamental biological or medical knowledge? What unanswerable question does this answer?

→ Clinical / Public Health Significance

How does this eventually benefit patients or populations? Even basic science proposals should connect to downstream human impact.

→ Economic / Policy Significance

For DBT and Wellcome Trust — how does this serve national priorities, industry sectors, or global health equity? These agencies weight this heavily.

→ Field-Level Significance

What new tools, methods, or frameworks will the field gain — even if the central hypothesis is wrong?

4

Approach — The Logical Progression Structure

Showing reviewers you have thought through everything

Most applicants write Approach as a list of experiments. Funded proposals write it as a logical argument — where each aim builds evidence, and the overall arc demonstrates rigour and feasibility.

The 5-Element Approach Structure (per Aim)**→ Rationale**

Why are you doing this aim? What evidence supports the logic of this experimental approach?

→ Experimental Design

What specifically will you do? Methods, sample sizes, timepoints, controls. Vague methods score down Approach.

→ Expected Outcomes

What do you expect to find — and why? Connect expected outcomes directly to your central hypothesis.

→ Potential Problems & Alternatives

What could go wrong? What will you do if it does? Showing you have anticipated problems signals scientific maturity. This is the section most applicants skip and reviewers most notice.

→ Interpretation

How will you interpret positive and negative results? A single negative result should not sink your entire proposal.

5

Budget Justification That Doesn't Kill Your Score

Every line is a scientific argument, not a financial one

The Scientific Budget Framework

→ Personnel first

Justify each person by their role in specific aims, not their job title. 'Dr X will lead Aim 2 genomic analysis (30% effort)' not 'Dr X — salary: \$X.'

→ Equipment by experiment

Every equipment cost should reference the specific experiment. 'Fluorescence microscope required for live cell imaging in Aim 1B' not 'equipment: \$X.'

→ No round numbers

Round numbers (\$5,000, \$10,000) signal estimates, not calculations. Use specific figures — \$4,840, \$9,360 — even if you arrive there by rounding internally.

→ Indirect costs explained

Briefly note your institution's indirect cost rate. Never leave indirect costs unexplained — reviewers sometimes query them.

PART 6 — AGENCY INTELLIGENCE

What Each Funding Body Actually Rewards

A

NIH (USA) — R01, R21, R03, SBIR/STTR

The world's largest biomedical funder — and the most competitive

NIH R01

\$1.5M avg

5 criteria: Significance, Investigator, Innovation, Approach, Environment (scored 1–9). Focus 60% of revision time on Approach — highest score weight and most reviewer variability. Your overall impact score is a weighted composite of all five.

NIH R21

\$275K avg

Exploratory/developmental mechanism. Reviewers expect earlier-stage science and accept higher risk. Preliminary data helpful but not expected to be comprehensive. R21s should not look like trimmed-down R01s.

SBIR/STTR

Up to \$2M

Innovation scored by commercial potential, not scientific novelty. You must answer: who will buy this, at what price, in what market? The Commercialisation Plan must be written by someone who understands biotech markets.

B

DBT & ICMR (India)

Mission-driven funding with a national priority framework

→ Connect your science explicitly to Make in India, Digital India, or national health priorities in the Significance section — even if the connection requires a sentence of explanation.

→ Show Indian institutional collaboration. Proposals with co-investigators at Indian universities are viewed more favourably than single-PI applications.

- Preliminary data from Indian patient populations significantly strengthens proposals — especially for ICMR.
- Budget justification must be meticulous — DBT review committees scrutinise budgets at a higher rate than NIH.
- Keep the Aims section clear and jargon-free — DBT review panels are multidisciplinary.

C

Wellcome Trust (UK/Global)

A values-driven funder with equity and global health at its centre

- Wellcome weighs global health equity heavily. Proposals addressing health challenges in low- and middle-income countries are strategically prioritised.
- The lay summary (or cover letter) is read by programme officers before the scientific committee. A compelling lay summary determines whether your proposal receives a sympathetic initial read.
- Wellcome funds people as much as projects. The Investigator section must make a compelling case for why you — specifically — are the right person for this work.
- Interdisciplinary proposals are actively encouraged. If your work crosses biological, social, or computational boundaries, frame this as a strength.
- Wellcome is comfortable with risk. A high-risk, high-reward proposal is not penalised — it is often rewarded. Do not hedge your hypothesis to make it sound safer than it is.

PART 7 — THE RESUBMISSION FORMULA

Turning a Rejection Into a Funded Grant

Most funded grants were rejected at least once. The difference between a failed resubmission and a funded one is almost always in the response to reviewers — not the science.

"Never argue with reviewers. Even when they are wrong. Every response is: acknowledge, address, and advance." — BioAI Lab Grant Strategy Team

The 4-Step Resubmission Formula

→ Read without emotion

Wait 48 hours after rejection. Read the summary statement as if it were written about someone else's proposal. Identify the 3 most substantive concerns — not the most hurtful ones.

→ Categorise each concern

Sort into: (a) scientific concerns addressable with new data, (b) misunderstandings resolvable with clarification, and (c) fundamental disagreements. The first two drive your revision.

→ Write the Introduction page

For NIH: one page to address reviewer concerns. Structure: here is what we changed and why — for every major concern. Never say a reviewer was wrong. Say 'we clarify that...' or 'we have added preliminary data showing...'

→ Revise strategically, not comprehensively

Targeted revisions score higher than wholesale rewrites. Reviewers notice when an applicant has responded precisely to their concerns. Major revisions to sections reviewers scored positively can lower your score.

\$18M+

Total Grants Secured

87%

First-Round Success Rate

120+

Proposals Submitted

42+

Countries Served

Ready to Win Your Next Grant?

BioAI Lab has helped clients secure \$18M+ in NIH, DBT, Wellcome Trust, NSF, SBIR/STTR and Horizon Europe grants — with an 87% first-round success rate. Every proposal is written by a PhD-level specialist matched to your exact research domain.

Book a free 30-minute grant consultation:

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